MIE1624: Introduction to Data Science, Analytics, and AI – Fall 2026

Assignment 1
Due Date: 11:59 pm, October 16th, 2026
Submit via Quercus

Background

For this assignment, you are responsible for answering the questions below based on the dataset provided. You are required to submit (only) two files. The first should be an IPython Notebook (.ipynb file) containing all the code of the analysis you performed to answer the questions. Please ensure that this notebook is not saved in any other formats, such as PDF or HTML. The second file is a 5-page report (PDF file) in which you will briefly describe what you have done and present the results of your analyses. In your report, you should use visual forms to present your results. How you decide to present your results (i.e., with tables/plots/etc.) is up to you, but your choice should make the results of your analysis clear and obvious.

Dataset

The 2025 Stack Overflow Developer Survey is a report on the state of software development. Now in its fifteenth year, Stack Overflow received over 49,000 responses from developers across 177 countries. The survey covers 62 questions spanning 314 different technologies, with new focus areas on AI agent tools, large language models (LLMs), and community platforms. More information on this survey can be found at: https://survey.stackoverflow.co.

The **survey_results_public.csv** dataset, available on both the mentioned website and Quercus, contains the survey results. The results from roughly 49,000 participants are shown in multiple columns, representing survey questions. Not all questions are answered by each participant, and responses contain various data types.

The original dataset has been cleaned for you and now contains fewer rows and columns, with rows containing null salaries removed. In the cleaned dataset provided for this assignment (**clean_stackoverflow_data.csv**), the second-to-last column ("ConvertedCompYearly") contains a numerical target variable, the annual salaries of the participants in USD. **You should work with the clean dataset for this assignment.**

Questions

The objective of this assignment is to explore the survey data to understand **(1)** distinctions between remote and hybrid job modes and **(2)** the effects of education on income level. The following tasks should be completed:

1. **[2 pts]** Perform Exploratory Data Analysis (EDA) to analyze the survey dataset and to summarize its main statistical characteristics. Present **3 meaningful figures** demonstrating Average Job Satisfaction (last column) by Job Title ("DevType" column), Mean Salary by Job Title, and Annual Salary vs. Age. Analyze each graph for anomalies and provide one sentence of diagnostic interpretation for each figure.

2. **[7 pts] Estimating Salary Differences Between Hybrid and Remote Job Modes**

   The job modes are listed in the 14th column of the cleaned dataset ("RemoteWork").

a. **[0.5 pts]** Compute and report descriptive statistics for the two groups of **hybrid** and **remote** job modes. Remove outliers if necessary and provide your rationale for either keeping or removing them.

b. **[1.5 pts]** Perform a two-sample t-test to compare average salaries between hybrid and remote job modes and report the test results. Explain what exactly your obtained p-value indicates.

   Note: First, perform a manual calculation using Python to compute the two-sample t-test statistic, including detailed step-by-step calculations of the means, variances, and pooled standard deviation for the two groups. Next, use Python's built-in functions to perform the t-test and compare the results with your manual calculations, discussing any discrepancies found between the two methods. You may need to check whether the assumptions required for the statistical testing are satisfied.

c. **[2 pts]** Bootstrap your data for comparing the mean salary for the two groups of hybrid and remote workers. Note that the number of instances you sample from each group should be relative to its size. Use 10,000 replications. Present your findings in three distinct figures: one demonstrating the two bootstrapped distributions (for hybrid and remote job modes), another depicting the distribution of the difference in means, and a third showcasing the standardized distribution of the difference in means.

d. **[1 pt]** Perform a two-sample t-test with a significance level of 0.05 threshold using bootstrapped data. Compare your results with those from 2.b and explain your findings.

e. **[1 pt]** Compute 95% confidence intervals (CIs) for the difference in mean salary between the hybrid and remote job-mode groups using two methods: (a) Formula-Based Approach (e.g., a t–distribution–based method) using the original data. (b) Bootstrap Approach using your previously generated bootstrap data. Finally, compare both sets of confidence intervals and discuss whether they reach the same conclusion about the difference in mean salaries. If they differ, explain possible reasons for these discrepancies.

f. **[1 pt]** In addition to comparing mean salaries, examine the median salaries of the two groups by performing a Mood's median test. Check if the results from the nonparametric test (i.e., Mood's p-value) and from bootstrapping (i.e., the two-sided p-value for the differences in bootstrapped medians) are close to each other. Discuss any notable similarities or differences.

3. **[3pts]** Select the education level column from the cleaned dataset ("EdLevel") and repeat steps **a** to **d**, but this time use Analysis of Variance (ANOVA) instead of a t-test for hypothesis testing to compare the mean salaries across three education groups: Bachelor's degree, Master's degree, and Professional degree. For the ANOVA, limit your analysis to survey participants from North America (i.e., the US and Canada). [**0.5pts** for **a; 0.5pts** for **b; 1.5pts** for **c; 0.5pts** for **d**].

   Note*:* You do not need to perform the ANOVA manually; using a built-in function is sufficient. Additionally, similar to question 2.c, question 3.c requires you to submit three figures: one demonstrating three bootstrapped distributions, another showing pairwise distributions of the difference in means, and a third depicting standardized distributions of the difference in means.

4. **[Bonus 2 pts] Client-Facing Interactive Dashboard** – note that this bonus can only make up for marks lost on Questions 1-3; it cannot raise your total assignment mark above 12 points

   Assume you work as a data analyst at a consultancy firm, and the analysis in Questions 1-3 was commissioned by one of your clients, a technology company reviewing its remote-work and hiring policies. You have completed the analysis, and instead of emailing static figures, you now want to send the client a single link to a professional dashboard that presents your findings. With the help of an LLM of your choice (e.g., Claude, ChatGPT) or a multi-agentic coding framework of your choice (Claude Code, OpenAI Codex, Cursor, etc.), build a visual, preferably interactive, dashboard that contains all of the figures and answers requested in this assignment: the three EDA figures, the descriptive statistics, the manual and built-in t-test results, the bootstrapped distributions and confidence intervals, the Mood's median test comparison, and the full ANOVA analysis for education level. Any mismatch between the dashboard and your notebook will result in no bonus marks. Organize the dashboard so that a non-technical client can follow it: open with a short executive summary of the key takeaways, then walk through each result with a plain-language explanation of what it means for the client. You are encouraged to be creative and may ask the LLM to add further insightful views beyond those required. You may deliver the dashboard in any form that supports sharing by link, such as a published Claude artifact, a standalone HTML page, or a Streamlit, Dash, or Hugging Face Spaces app. Include the public link on the first page of your report, and make sure it opens and works correctly without a login, in a private/incognito browser window, and remains accessible until final grades are released. A link that is broken, private, or requires an account to view will not be graded. Because the dashboard is public, embed only aggregated results or the summary data needed for your figures rather than the full dataset. The dashboard does not count toward the 5-page report limit. *Note that Questions 1-3 are worth 12 pts in total. Bonus Question 4 is worth up to 2 additional points, but these can only offset points lost in Questions 1-3, and your final assignment mark is capped at 12 pts regardless of the bonus.*

Submission

1) Produce a 5-page report (including figures) explaining your response to each question for the given dataset and detailing the analysis you performed.
Note: The report submission is strictly limited to 5 pages, including figures, explanations, and all other content. While this may seem challenging, part of this exercise is to cultivate the skill of presenting complex information concisely and clearly.

2) Produce a Jupyter (IPython) Notebook detailing the analysis you performed to answer the questions for the given dataset.

What to submit

1. Submit via Quercus a Jupyter (IPython) notebook containing your implementation and motivation for all the steps of the analysis with the following naming convention: **lastname_studentnumber_assignment1.ipynb**.

   Make sure that you **comment** on your code appropriately and describe each step in sufficient detail. Respect the above convention when naming your file, making sure that all letters are lowercase, and underscores are used as shown. **A program that cannot be evaluated because it varies from**

specifications will receive zero marks. Also, to receive the marks, your code must run without errors.

2. Submit a report in PDF, including the findings from your analysis. Use the following naming conventions **lastname_studentnumber_assignment1.pdf**. **Page size: US Letter (8.5"x11"), 1-inch margin, font size: 11 pt.**

Late submissions will receive a standard penalty:

- up to one hour late - no penalty
- one day late - 15% penalty
- two days late - 30% penalty
- three days late - 45% penalty
- more than three days late - 0 mark

Tools

- **Software:**
  - **Python Version 3.X** is required for this assignment. All libraries are allowed but here is a list of the major libraries you might consider: Numpy, Scipy, Sklearn, Matplotlib, Pandas. **Specify `!pip install <library_name>` commands in the first cell of your notebook for all the 3rd party libraries you use. Make sure your Jupyter notebook runs in Google Colab.**
  - No other tool or software besides Python and its component libraries can be used to touch the data files. For instance, using Microsoft Excel to clean the data is NOT allowed.
  - Read the required data file from the same directory as your notebook: for example, pd.read_csv("clean_stackoverflow_data.csv").

- **Required data files:**
  - **clean_stackoverflow_data.csv**: survey responses with yearly compensation.
  - The data file cannot be altered by any means. The Jupyter notebook will be run using a local version of this data file. Do not save anything to file within the notebook and read it back.

Other requirements

1. You may decide to complete parts of this assignment **through an LLM** such as ChatGPT, Claude, Gemini, etc. Similarly, you may use **multi-agent coding systems** such as Claude Code, OpenAI Code, Gemini CLI, Copilot, and Cursor to generate Python code and the report. In that case, in the appendix of your report, reference which LLM(s) and/or multi-agentic system(s) you have utilized and provide samples of the **prompts** you used to elicit answers, code, and report with suggested figures. If the LLM's initial responses contained errors or required verification, explain how you **validated and corrected** those (reflect on how prompt engineering and iterative feedback helped improve the final output).
2. A large portion of marks is allocated to analysis and justification. Full marks will not be given for code alone.

3. Output must be shown and readable in the notebook. The only files that can be read into the notebook are those posted for the assignment, without modification. All work must be done within the notebook.
4. The notebook should be presentable; do not show large amounts of raw output.
5. Ensure the code runs in full before submitting. Just before you submit, rerun the entire notebook (navigate to Kernel => Restart Kernel and Run all Cells). Ensure that there are no errors.

"The highest activity a human being can attain is learning for understanding, because to understand is to be free."
— Baruch Spinoza

Best of Luck!